

Notes: Keys, Mahoney & Yang (RFS, 2023)
What Determines Consumer Financial Distress? Place- and Person-Based Factors

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1 Big picture

- **Question.** What explains large and persistent geographic differences in consumer financial distress: local place-based factors or persistent person-based characteristics?
- **Data.** The paper uses a national TransUnion credit-report panel tracking roughly 35 million individuals monthly over 2000–2016.
- **Outcomes.** The main distress measures are debt in collections, credit card delinquency, Chapter 7 bankruptcy, and Chapter 13 bankruptcy.
- **Descriptive fact.** The Deep South has much higher distress than the Upper Midwest. At the CZ level, debt in collections is 37.3% in the Deep South versus 19.9% in the Upper Midwest.
- **Research design.** The movers design studies how outcomes change when individuals move to places with different distress rates.
- **Main result.** Debt in collections and credit card delinquency converge by less than 10%, so person-based factors explain most of the geographic variation.
- **Bankruptcy contrast.** Bankruptcy filing rates converge much more: roughly 27% for Chapter 7 and 33% for Chapter 13.
- **Economic takeaway.** Routine distress outcomes are mostly person-based, but bankruptcy is substantially place-based because it is mediated by local legal institutions, lawyers, and legal culture.

2 Data and measurement

2.1 Credit report data

The paper uses monthly TransUnion credit reports from 2000–2016. The panel begins as a random 10% sample of individuals with TransUnion credit records in 2000 and is refreshed monthly with new credit-report entrants. The average raw month contains 35.6 million individuals; after dropping missing ages and ages below 20 or above 80, the average analysis sample contains about 30.1 million individuals.

Interpretation: The panel is large enough to estimate local distress rates, follow movers precisely, and study rare outcomes such as bankruptcy.

2.2 Financial distress outcomes

- **Debt in collections:** indicators and balances for new collections in the past 3 years and stock collections measures, separated into medical and nonmedical collections when possible.
- **Credit card delinquency:** one or more credit cards 30+ days past due in the past 3 years.
- **Bankruptcy:** any filing, Chapter 7 filing, and Chapter 13 filing in the past 3 years, measured per 1,000 individuals.

Interpretation: Three-year windows smooth noise and make rare outcomes comparable across places.

2.3 Geographic units and movers sample

The descriptive analysis uses commuting zones (CZs). The movers analysis focuses on individuals observed for the full sample window, ages 30–80 in the last period, with exactly one across-CZ move between 2004 and 2007. The baseline sample contains 145,805 movers.

Interpretation: Restricting moves to 2004–2007 gives at least 4 years of premove data and 8 years of post-move data, helping test pre-trends and study medium-run convergence.

2.4 Size of move

For mover i , the size of the move is

$$\hat{\delta}_i = \bar{y}_{nonmovers}^{dest(i)} - \bar{y}_{nonmovers}^{origin(i)}.$$

A positive value means moving to a more distressed place; a negative value means moving to a less distressed place.

Interpretation: The size of move is a continuous treatment intensity. It measures how different the destination is from the origin for a given distress outcome.

3 Models and empirical specifications

3.1 Variance decomposition

$$\sum_i \sum_t (x_{it} - \bar{x})^2 = \sum_i \sum_t (x_{it} - \bar{x}_i)^2 + \sum_i T(\bar{x}_i - \bar{x})^2.$$

The first term is within-place time-series variation. The second term is persistent cross-sectional variation across CZs.

Interpretation: The variance decomposition motivates the movers design: most variation is across places, so the paper asks whether those place differences are caused by places or by people who live there.

3.2 Event-study specification

$$y_{it} = \alpha_i + \alpha_y + \alpha_q + \alpha_r + \sum_{r \neq -1} \theta_r \hat{\delta}_i \mathbf{1}\{eventtime = r\} + x'_{it} \beta + \varepsilon_{it}.$$

Here α_i are individual fixed effects, α_y calendar-year fixed effects, α_q calendar-quarter fixed effects, and α_r event-time fixed effects. The omitted period is the last quarter before the move.

Interpretation: $\theta_r = 0$ means no convergence to the destination; $\theta_r = 1$ means full convergence. Intermediate values measure partial place effects.

3.3 Robust specifications

- Same-origin specification: includes origin CZ $\times$ event-time fixed effects.
- Same-destination specification: includes destination CZ $\times$ event-time fixed effects.
- Alternative size-of-move definitions: ZIP $\times$ age and county-level nonmover averages.
- Bankruptcy robustness: allows the size of move to vary around the 2005 bankruptcy reform.

Interpretation: These specifications address concerns that movers select into destinations because of persistent origin or destination shocks.

4 Identification and empirical design

4.1 Core identification assumption

The identifying assumption is a parallel-trends condition: conditional on controls, the size of the move is uncorrelated with differential changes in financial distress that would have occurred even without the move.

Interpretation: If movers to high-distress places were already becoming more distressed before moving, the estimated place effects would not be credible.

4.2 Why movers identify place effects

Person-based characteristics move with the individual. Place-based factors are tied to the destination. If financial distress converges to the destination after a move, that convergence is evidence of a place effect.

Interpretation: The design separates portable individual characteristics from nonportable local institutions and environments.

4.3 Pre-trend validation

The paper tests whether premove outcome trends are related to the size of the move. Figure 2 shows little relationship, supporting the identifying assumption.

Interpretation: The absence of pre-trends is central: it shows that movers to high-distress destinations were not already trending toward higher distress before moving.

4.4 Same-origin and same-destination tests

The same-origin specification compares people leaving the same origin for destinations with different distress levels. The same-destination specification compares people arriving at the same destination from origins with different distress levels.

Interpretation: These tests reduce bias from origin-specific shocks and destination-specific shocks.

4.5 What the design cannot identify

The design estimates the relative importance of place and person factors, but it does not fully identify the mechanisms inside the person component. Person effects may include financial literacy, income volatility, wealth, risk preferences, default stigma, human capital, or early-life place exposure.

Interpretation: The design is a decomposition tool, not a complete structural model of consumer financial distress.

5 Tables and figures

5.1 Table 1: Financial distress measures; Table 2: Variance decomposition

Read: Table 1 shows that 28.4% of individuals have a new collection in the past 3 years, 34.1% have debt in collections as a stock measure, and 10.1% have credit card delinquency in the past 3 years. Table 2 shows that most variation in distress is cross-sectional across CZs rather than within-CZ over time.

Detailed interpretation: Table 1 establishes the prevalence of distress. Table 2 justifies the paper’s cross-sectional geographic focus. Chapter 13 has an especially large cross-sectional component, consistent with local legal institutions being important.

Economic message: The empirical problem is not only cyclical distress; it is persistent geographic inequality in distress.

Table 1
Financial distress measures

	Mean	SD	Median	Pct 75	Pct 90	Pct 95	Pct 99
Collections - Flow							
Debt in collections in past 3 years (%)	28.4	45.1	0	100	100	100	100
Balance of collections in past 3 years (\$)	818.0	3,192.7	0	237	2,000	4,121	13,386
Collections - Stock							
Debt in collections (%)	34.1	47.4	0	100	100	100	100
Medical collections (%)	22.1	41.5	0	0	100	100	100
Nonmedical collections (%)	24.2	42.8	0	0	100	100	100
Collections balance (\$)	1,350.9	4,843.2	0	606	3,401	6,748	21,234
Medical collections balance (\$)	697.3	3,448.4	0	0	1,245	3,178	13,770
Nonmedical collections balance (\$)	624.4	2,577.0	0	0	1,445	3,218	10,993
Credit Card							
Credit card delinquency in past 3 years (%)	10.1	30.1	0	0	100	100	100
Credit card delinquency in past 3 years (% cond'l)	14.1	34.8	0	0	100	100	100
Bankruptcy							
Bankruptcy filings in past 3 years (per 1,000)	14.8	120.9	0	0	0	0	1,000
Chapter 7 filings in past 3 years (per 1,000)	10.5	101.8	0	0	0	0	1,000
Chapter 13 filings in past 3 years (per 1,000)	4.5	67.2	0	0	0	0	0
N	29,398,845						

This table presents summary statistics for measures of financial distress constructed using a 10% random sample of TransUnion credit records from June 2015. Debt in collections measures are indicators for 1+ debt in collections. We include both flow measures of debt in collections over the past 3 years and the stock measures of whether the individual has 1+ debt in collections. Credit card delinquency is an indicator for 1+ credit card that is 30+ DPD over the past 3 years. Bankruptcy filings are the number of individuals out of 1,000 who file for any bankruptcy, Chapter 7, and Chapter 13, respectively, in the last 3 years. See Section 1 for more details on variable construction.

(a) Table 1: Financial distress measures. Summary statistics for collections, delinquency, and bankruptcy.

Table 2
Variance decomposition

	Time-series	Cross-section
Debt in collections in past 3 years (%)	35.19	64.81
Debt in collections (%)	27.77	72.23
Medical collections (%)	22.87	77.13
Nonmedical collections (%)	14.04	85.96
Credit card delinquency in past 3 years (%)	57.37	42.63
Chapter 7 filings in past 3 years (per 1,000)	43.00	57.00
Chapter 13 filings in past 3 years (per 1,000)	10.54	89.46

This table presents a decomposition financial distress into its time-series and cross-sectional components, based on a panel data set of average financial distress for each CZ $\times$ year from 2000 to 2016.

(b) Table 2: Variance decomposition. Time-series versus cross-sectional components of distress variation.

5.2 Figure 1: Geographic variation in financial distresses; Table 3: Geographic variation in financial distress

Read: Figure 1 maps CZ-level distress. Table 3 quantifies the map: debt in collections is 37.3% in the Deep South versus 19.9% in the Upper Midwest; Chapter 13 filings are 10.8 versus 2.8 per 1,000.

Detailed interpretation: The Chapter 7/Chapter 13 split is revealing: Chapter 13 is much higher in the Deep South, while Chapter 7 is lower. This suggests bankruptcy chapter choice is shaped by local legal culture and lawyer networks rather than just borrower hardship.

Economic message: The large regional differences motivate the movers design.

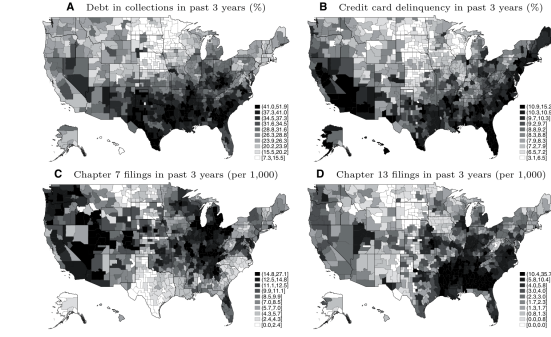


Figure 1
Geographic variation in financial distress
This figure shows CZ-level maps of financial distress. CZ means are constructed using a 10% random sample of TransUnion credit records from June 2015. Debt in collections is an indicator for 1+ debt in collections in the last 3 years. Credit card delinquency is an indicator for 1+ credit card that is 30+ DPD in the last 3 years. Bankruptcy filings are the number of individuals out of 1,000 who file Chapter 7 and 13, respectively, in the last 3 years. See Section 1 for more details on variable construction.

Table 3

Geographic variation in financial distress

	(1) Mean	(2) SD	(3) 75%–25%	(4) 90%–10%	(5) Deep South	(6) Upper Midwest	(7) Deep South- Upper Midwest
Collections - Flow							
Debt in collections in past 3 years (%)	28.4	7.1	11.0	18.7	37.3	19.9	17.4
Balance of collections in past 3 years (\$)	818.0	286.4	411.9	719.3	1,117.5	601.3	516.2
Collections - Stock							
Debt in collections (%)	34.1	7.9	11.5	20.8	43.5	23.8	19.8
Medical collections (%)	22.1	8.1	11.5	22.5	31.6	14.4	17.2
Nonmedical collections (%)	24.2	5.2	7.1	14.1	29.1	17.4	11.7
Collections balance (\$)	1,350.9	461.4	580.4	1,160.5	1,723.2	987.5	735.7
Medical collections balance (\$)	697.3	373.4	500.8	962.0	1,064.8	541.9	522.9
Nonmedical collections balance (\$)	624.4	188.7	303.4	483.8	627.0	423.0	204.0
Credit Card							
Credit card delinquency in past 3 years (%)	10.1	1.5	2.0	3.4	10.7	7.2	3.5
Credit card delinquency in past 3 years (% cond'l)	14.2	2.6	3.5	6.7	17.1	9.3	7.7
Bankruptcy							
Bankruptcy filings in past 3 years (per 1,000)	14.8	6.4	8.7	17.8	19.8	14.5	5.3
Chapter 7 filings in past 3 years (per 1,000)	10.5	4.6	7.5	12.1	9.2	11.9	–2.6
Chapter 13 filings in past 3 years (per 1,000)	4.5	4.1	2.9	7.7	10.8	2.8	8.1

This table presents statistics on CZ-level measures of financial distress. CZ means are constructed using a 10% random sample of TransUnion credit records from June 2015. Summary statistics are calculated using the CZ-level data, with CZs weighted by the number of individual-level observations in each CZ so means are representative of the underlying individual-level data. The Deep South is defined as Alabama, Arkansas, Georgia, Louisiana, Mississippi, and South Carolina and Upper Midwest as Iowa, Minnesota, North Dakota, South Dakota, Wisconsin, and the upper peninsula of Michigan. See Table 1 for more details on the financial distress measures.

5.3 Figure 2: Premove change in financial distress by size of move; Figure 3: Pre- vs. post-change in financial distress by size of move

Read: Figure 2 tests whether premove changes are related to the size of the move. Figure 3 shows that post-move changes are positively related to the size of the move.

Detailed interpretation: The pair of figures is the visual foundation of the research design: no pre-trend before the move, but convergence after the move.

Economic message: The patterns support the view that movement to a new place changes outcomes rather than simply reflecting prior trends.

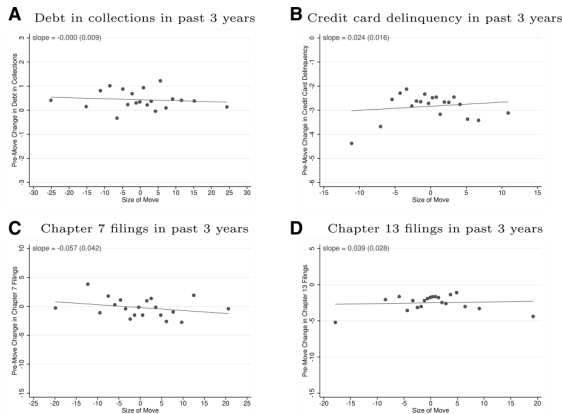


Figure 2

Premove change in financial distress by size of move

This figure shows binned scatter plots of the *pre-move* change in the outcome against the size of the move δ_i . The vertical axis shows the average financial distress 1 year premove minus the average financial distress 3 years premove. The horizontal axis shows δ_i , the average financial distress between the destination and origin ZIP codes. The data are split by ventiles of δ_i , and each point represents the average in that bin. The plots also show the line of best fit, estimated using the underlying data, and its slope and standard error.

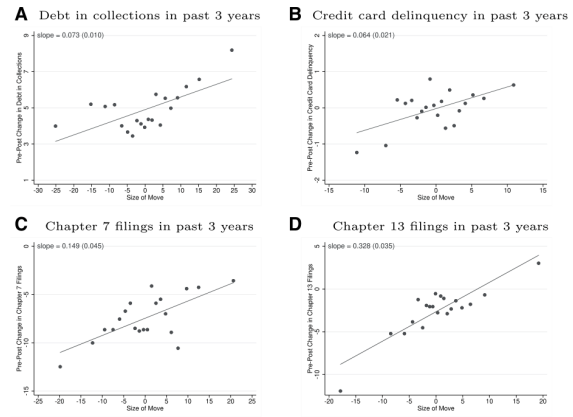


Figure 3

Pre- vs. post-change in financial distress by size of move

This figure shows binned scatter plots of the *pre- vs. post-* change in the outcome against the size of the move δ_i . The vertical axis shows the average financial distress 3 years post-move minus the average financial distress 3 years premove. The horizontal axis shows δ_i , the average financial distress between the destination and origin ZIP codes. The data are split by ventiles of δ_i , and each point represents the average in that bin. The plots also show the line of best fit, estimated using the underlying data, and its slope and standard error.

(a) Figure 2: Premove change in financial distress by size of move. Pre-trend diagnostic.

(b) Figure 3: Pre- vs. post-change in financial distress by size of move. Initial convergence evidence.

5.4 Figure 4: Event-study plots; Table 4: Event-study estimates

Read: Figure 4 shows event-study convergence. Collections and credit card delinquency converge by less than 10%; Chapter 7 and Chapter 13 converge much more. Table 4 estimates baseline 6-year place effects of 0.2665 for Chapter 7 and 0.3298 for Chapter 13.

Detailed interpretation: The table compresses the dynamic patterns into medium-run estimates and shows robustness to same-origin, same-destination, ZIP-by-age, and county-level definitions.

Economic message: Bankruptcy is much more place-sensitive than routine default outcomes.

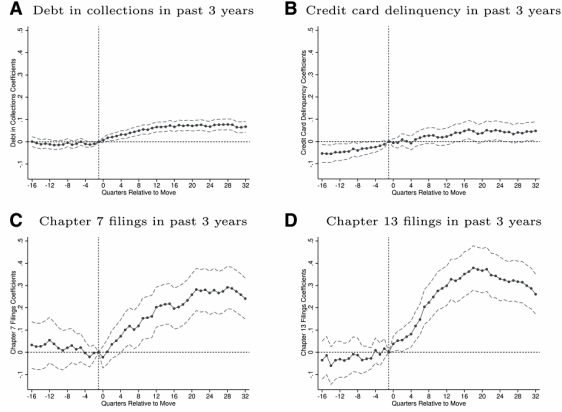


Figure 4
Event-study plots
The figure shows place-based effects, θ_t , from event study regressions of financial distress on the size of the move δ_t , individual and time fixed effects, and other controls. The dashed lines represent 95% confidence intervals, based on standard errors clustered by origin $\times$ destination CZ.

Table 4
Event-study estimates

Financial distress measures	Fraction converged to outcome at destination 4 years post-move				
	(1) All movers	(2) Same origin	(3) Same destination	(4) ZIP $\times$ age level	(5) County level
Debt in collections 3 years	0.0681 (0.0122)	0.0358 (0.0123)	0.0487 (0.0125)	0.0622 (0.0105)	0.1615 (0.0173)
Debt in collections	0.0788 (0.0095)	0.0536 (0.0095)	0.0611 (0.0097)	0.0659 (0.0081)	0.1539 (0.0141)
Medical collections	0.2042 (0.0130)	0.1291 (0.0131)	0.1908 (0.0133)	0.1575 (0.0110)	0.3036 (0.0164)
Nonmedical collections	0.0409 (0.0120)	0.0213 (0.0122)	0.0337 (0.0123)	0.0490 (0.0103)	0.0956 (0.0187)
Credit card delinquency 3 years	0.0473 (0.0215)	0.0681 (0.0215)	0.0149 (0.0213)	0.0326 (0.0157)	0.0710 (0.0376)
Chapter 7 lings 3 years	0.2665 (0.0500)	0.2088 (0.0501)	0.2212 (0.0509)	0.0863 (0.0325)	0.5068 (0.0897)
Chapter 13 lings 3 years	0.3298 (0.0503)	0.3578 (0.0522)	0.2095 (0.0535)	0.1097 (0.0422)	0.4940 (0.0625)

This table shows the place-based effects θ_t at 6 years (24 quarters) after the move from event study regressions of financial distress on the size of the move δ_t , individual and time fixed effects, and other controls. Standard errors, clustered by origin $\times$ destination CZ, are shown in parentheses. Column 1 shows the baseline specification. In column 2, we isolate variation from moves “from the same place” by including fully interacted origin CZ $\times$ event time fixed effects. In column 3, we isolate variation from moves “to the same place” by including fully interacted destination CZ $\times$ event time fixed effects. In columns 4 and 5, we show alternative specifications where we construct δ_t using nonmovers in the movers’ origin and destination ZIP code $\times$ 10-year age bin, and using nonmovers in the movers’ origin and destination county.

(a) Figure 4: Event-study plots. Place-effect coefficients by event time.

(b) Table 4: Event-study estimates. Medium-run place effects and robustness specifications.

5.5 Figure 5: Event-study plots: Debt in collections stock measures

Read: Figure 5 decomposes collections into stock, medical, and nonmedical measures. Medical collections have a place effect of roughly 20%, while nonmedical collections have a much smaller effect of about 4%.

Detailed interpretation: The medical result is consistent with local hospitals, doctors, and debt collectors varying in collection practices.

Economic message: Place effects are concentrated where local institutions directly mediate the path from unpaid bills to credit-report distress.

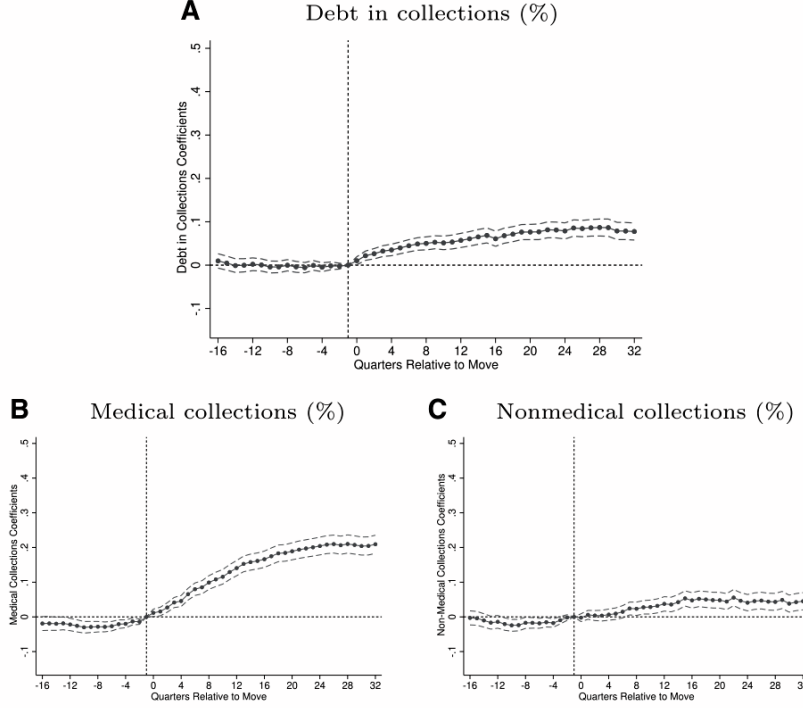


Figure 5
Event-study plots: Debt in collections stock measures
The panels show place-based effects θ_r from event study regressions of financial distress on the size of the move δ_i , individual and time fixed effects, and other controls. The dashed lines represent 95% confidence intervals, based on standard errors clustered by origin $\times$ destination CZ.

Figure 5: Event-study plots: Debt in collections stock measures. Stock, medical, and nonmedical collections.

6 Short summary

- The paper studies geographic variation in consumer financial distress using a massive TransUnion panel.
- Most variation is cross-sectional across commuting zones.
- The movers design estimates how much outcomes converge to destination distress rates.
- Collections and credit card delinquency show little convergence, so person-based factors dominate.
- Medical collections show a larger place effect, suggesting local billing and collection practices matter.
- Bankruptcy shows large place effects, especially Chapter 13.
- The core result is outcome-specific: places matter for bankruptcy, but people matter more for routine distress.

7 Conclusion

7.1 Core conclusion

Keys, Mahoney, and Yang show that the determinants of consumer financial distress differ by outcome. Collections and credit card delinquency are mostly person-based, while bankruptcy is substantially place-based.

7.2 Economic intuition

Moving separates portable person characteristics from destination-specific environments. If an outcome converges to the destination, local institutions matter. If it does not, persistent individual characteristics dominate.

7.3 Interpretation of asymmetric results

The small convergence for collections and delinquency suggests that local credit-market conditions and state laws are not the main cause of those disparities. The larger bankruptcy convergence suggests that legal institutions, lawyers, exemptions, and local filing norms play a major role.

7.4 Policy implications

Policies focused on local credit supply may have limited effect on collections and delinquency disparities. For those outcomes, interventions may need to target persistent person-level factors such as financial literacy, wealth buffers, income volatility, or early-life exposure. Bankruptcy outcomes may be more responsive to legal access, filing costs, attorney networks, and information provision.

7.5 Research implications

The paper leaves open what generates the person-based component. Future research should identify whether financial literacy, default stigma, risk preferences, wealth, income volatility, or early-life neighborhood effects are the most important mechanisms.

7.6 Takeaway

The paper reframes financial distress as a place-person decomposition. The correct policy response depends on the margin: routine delinquency is mostly portable with the person, while bankruptcy is strongly shaped by place.